

Endpoint construction

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May 26, 2026

0.1 Endpoint construction

The endpoint is deliberately asymmetric across the two data sources. The randomized component (ACTG 175) is used as published: a single composite indicator and one follow-up time, with the underlying components not separately released. The observational component (MACS) is reconstructed component-wise from the public release and then assembled to match a chosen target endpoint. Five candidate assemblies are tabulated below. The Kaplan–Meier diagnostic figure uses variant v5 (overall survival) for signal quality, while the publication-target FusionForest fit uses variant v3 (*EFS-cd4-fix*), the closest analogue of the ACTG 175 composite.

ACTG 175 (RCT) side

ACTG 175 releases each subject’s endpoint pre-assembled as a pair (`days`, `cens`), where `cens` $\in \{0, 1\}$ is the composite indicator for the first occurrence of any of (i) confirmed $\geq 50\%$ decline in CD4 count from baseline, (ii) AIDS-defining diagnosis, or (iii) death from any cause; and `days` is the time from randomization to that event or to administrative censoring, with a maximum follow-up of approximately 1,231 days. The three components are not released separately, so the RCT side cannot be respecified as a subset of the composite. On the analysis scale we work with $T = \text{days}/365.25$ in years and $\delta = \text{cens}$.

A separate CD4-50% reconstruction is also computed from the trial release (`cd_dec1_50`, `cd_dec1_days`) by comparing the 20- and 96-week CD4 measurements to half of baseline and approximating the event time by the nominal window midpoint (140 days for the 20-week visit, 672 days for the 96-week visit). This is reported as a diagnostic only: it is unconfirmed, ignores AIDS and death, and is never used as the analysis endpoint.

MACS (RWD) side: per-subject components

The MACS base frame, constructed once per analysis, carries every per-subject component as either a calendar year or a binary flag, independent of which variant is later assembled. The components and their sources are summarized in Table 1.

Two construction choices are non-trivial.

CD4 reconstruction. The ACTG 175 $\geq 50\%$ -decline rule does not match the `SELFCD4D` field in MACS, which encodes an absolute sub-threshold crossing (< 200 cells/ μL or $< 14\%$). We therefore rebuild the relative rule per subject from the `lab_rslt` time series: take the earliest

Field	Source	Meaning
<code>anchor_year</code>	<code>min(AVQY)</code> in 1991–95 with <code>AVNW = 2</code>	time-zero (year resolution)
<code>aids_yr</code>	<code>outcome.DATE1yy</code> if <code>AIDSCASE ∈ {2, 3}</code>	year of CDC AIDS diagnosis
<code>cd4_abs_yr</code>	<code>outcome.SELFCD4Dyy</code> if <code>> 0</code>	year of first absolute <code>CD4 < 200</code> or <code>< 14%</code>
<code>cd4_rel_yr</code>	first <code>LDATY</code> with <code>LEU3N ≤ 0.5 · cd4_base</code>	year of first relative <code>≥ 50%</code> <code>CD4</code> decline
<code>cd4_base</code>	first 1991–95 <code>LEU3N</code> per subject	baseline <code>CD4</code> for the relative rule
<code>death_any</code>	<code>DEATH ∈ {1, 2, 3, 4}</code>	all-cause death indicator (date redacted)
<code>death_aids</code>	<code>DEATH ∈ {1, 2}</code>	AIDS-related death indicator (date redacted)
<code>last_year</code>	<code>max(LDATY)</code> in <code>lab_rslt</code>	last contact year; death-time fallback / censoring

Table 1: MACS per-subject component fields. Years are calendar years extracted from the public release; flags are derived from the `outcome` form. `LEU3N` is the `lab_rslt` `CD4` count, `LDATY` the lab visit year, `AVQY` the drug-form visit year, and `AVNW` the “currently taking” indicator on a drug record.

1991–95 `LEU3N` as `cd4_base`, then take `cd4_rel_yr` as the year of the first subsequent lab with `LEU3N ≤ 0.5 · cd4_base`. This reconstruction is a one-shot threshold crossing rather than a confirmed decline, which is a known asymmetry against the RCT-side endpoint and is sensitive to non-uniform `CD4`-measurement frequency between subjects.

Missing death dates. The field `DTHDATEyy` is redacted from the public release. We therefore retain death indicators (`death_any`, `death_aids`) but no death dates. For subjects flagged as deceased without a preceding AIDS or `CD4` year, the event-time fallback is the subject’s `last_year`, i.e. the year of their last laboratory visit. This is a downward bias on death-event times that propagates into the upper tail of the survival distribution.

A prior-AIDS exclusion is applied as a final step: subjects with `aids_yr < anchor_year` are dropped so that the cohort is AIDS-free at time-zero, matching the trial’s entry criterion.

Assembly

Given the base frame and a boolean specification of which components feed the composite, the assembler `combine_macs(·)` computes

$$\begin{aligned}
\text{ev_yr} &= \min\{\text{cd4_abs_yr}, \text{cd4_rel_yr}, \text{aids_yr}, \text{last_year} \cdot \mathbf{1}\{\text{death active}\}\}_{\text{activated}}, \\
\text{end_year} &= \begin{cases} \text{ev_yr} & \text{if any activated component fires,} \\ \text{last_year} & \text{otherwise (right-censored at last contact),} \end{cases} \\
T &= \max(\text{end_year} - \text{anchor_year}, 0.5), \\
\delta &= \mathbf{1}\{\text{any activated component fires}\}.
\end{aligned}$$

Non-fired components contribute $+\infty$ to the rowwise minimum and so are ignored. Death contributes its fallback year (`last_year`) to the minimum only when the chosen death flag is active and the subject is flagged dead; otherwise the death component does not enter at all. The half-year floor on T accommodates year-resolution event times when event and anchor coincide, where the elapsed time would otherwise round to exactly zero and trigger a $\log(0)$ on the log-survival scale used downstream.

Variant menu

Five composite specifications are tabulated to map the activation choices onto recognizable named endpoints (Table 2).

Variant	Name	Components combined (event = earliest activated year)
v1	EFS-current	absolute CD4 + AIDS + AIDS-related death
v2	EFS-death-fix	absolute CD4 + AIDS + all-cause death
v3	EFS-cd4-fix	relative CD4 + AIDS + all-cause death
v4	PFS	AIDS + all-cause death
v5	OS	all-cause death

Table 2: MACS endpoint variants. v1 is the original specification; v2 corrects a death-code restriction that silently dropped roughly a quarter of recorded deaths by keeping only AIDS-related death codes; v3 swaps the absolute CD4 component for the confirmed-decline analogue and is the closest match to the ACTG 175 composite. v4 and v5 are progressively narrower endpoints used for sensitivity.

Which variant the diagnostic figure uses

The Kaplan–Meier diagnostic figure does not call the variant menu directly: it reimplements variant v5 (OS) inline. The event is `DEATH` $\in \{1, 2, 3, 4\}$, the event time is `last_year` (death-date fallback, since no death date is recorded), and the censoring time for survivors is also `last_year`. Elapsed time is $T = \max(\text{last_year} - \text{anchor_year}, 0.5)$. The figure labels this curve “EFS” for visual comparison with the RCT side, but mechanically it is overall survival — a strict subset of the RCT composite, chosen because death is the only event MACS captures reliably enough to give a clean survival signal at year resolution.

Time computation, side by side

Table 3 summarizes the time fields on the two sides.

Quantity	ACTG 175	MACS
Anchor	date of randomization (day)	$\min(\text{AVQY})$ in 1991–95 with $\text{AVNW} = 2$ (year)
Event time	<code>days</code> since randomization	rowwise min over activated component years
Censoring	published in <code>days</code>	<code>last_year</code> (year of last lab visit)
Analysis T	<code>days/365.25</code>	$\max(\text{end_year} - \text{anchor_year}, 0.5)$
Resolution	day	year (with 0.5-year floor)

Table 3: Time-axis construction on the two sides.

The 0.5-year floor on the MACS side biases short-fuse events upward by up to one year. It is the most visible footprint of the year-resolution public release on the analysis; its impact concentrates on subjects whose event year equals their anchor year and is largest for cohorts enriched in fast events.

Cohort definition versus endpoint

The cohort definition (which subjects enter the at-risk set) is logically distinct from the endpoint construction (how each subject's event and time are defined). Changing the cohort — for example, restricting to incident users of the contrast regimens within 1991–95 by excluding any subject with a pre-1991 record of $AVNW = 2$ on a regimen in the contrast code set — leaves the endpoint mechanics unchanged. The same event field on a different at-risk set yields different event counts, different follow-up distributions, and therefore different survival curves, without any change to the assembly logic above.